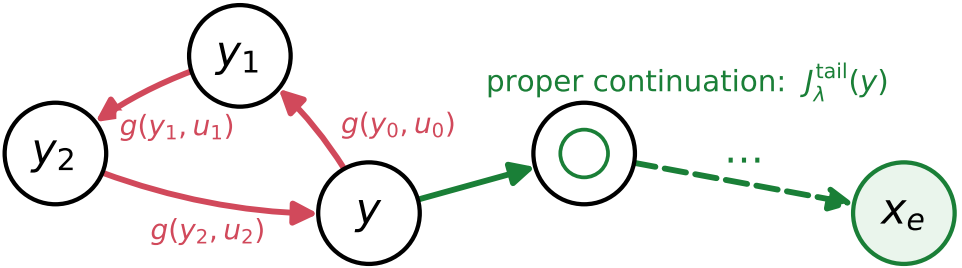
cycle C ($\ell = 3$), repeat forever:

value $= \dfrac{g_{\text{loop}}(C)}{1-\lambda^{\ell}}$



improper optimum at $y \iff g_{\text{loop}}(C) < (1-\lambda^{\ell}) J_\lambda^{\text{tail}}(y)$